

The Dot Product

ANSWER KEY



Computing dot products and angles

1 Let $\mathbf{u} = \langle 2, 5 \rangle$ and $\mathbf{v} = \langle -3, 4 \rangle$. Find:

a $\mathbf{u} \cdot \mathbf{v} \quad (2)(-3) + (5)(4) = 14.$

b $|\mathbf{u}|$ and $|\mathbf{v}| \quad |\mathbf{u}| = \sqrt{29}, |\mathbf{v}| = 5.$

c the angle between $\mathbf{u}$ and $\mathbf{v}$, to one decimal place $\cos \theta = \frac{14}{5\sqrt{29}} \approx 0.5199$, so $\theta \approx 58.7^\circ$.

2 Determine whether each pair of vectors is orthogonal:

a $\langle 4, -6 \rangle$ and $\langle 3, 2 \rangle \quad 12 - 12 = 0$: orthogonal. b $\langle 1, 2 \rangle$ and $\langle 2, 1 \rangle \quad 2 + 2 = 4 \neq 0$: not orthogonal.

3 Find the value of k so that $\langle k, 3 \rangle$ is orthogonal to $\langle 2, -4 \rangle$.

$2k - 12 = 0$, so $k = 6$.

Projections and work

4 Find the vector projection of $\mathbf{u} = \langle 6, 2 \rangle$ onto $\mathbf{v} = \langle 3, 0 \rangle$.

$\text{proj}_{\mathbf{v}} \mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|^2} \mathbf{v} = \frac{18}{9} \langle 3, 0 \rangle = \langle 6, 0 \rangle.$

5 A force $\mathbf{F} = \langle 8, 5 \rangle$ N moves an object along the displacement $\mathbf{d} = \langle 10, 0 \rangle$ m. Find the work done.

$W = \mathbf{F} \cdot \mathbf{d} = 80$ J.